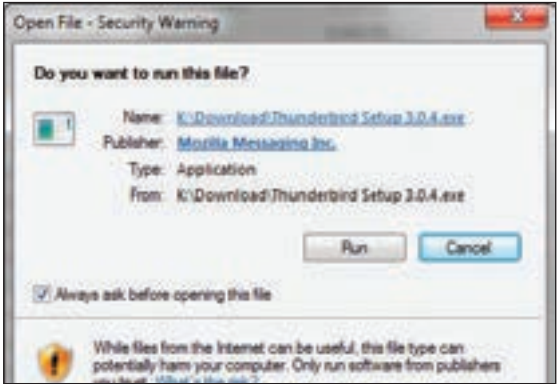


access all the streams of a file.

You may have noticed how applications that have been downloaded from the internet display a warning dialog on launching. The dialog may look something like the following:



Ads launch security warning

The reason this happens is that Windows applications that download files from an external server add metadata to the downloaded file that tells Windows that it has been downloaded from a third party. On finding this stream, Windows warns you of the source of the file before launching it. This metadata is stored in an alternate data stream of the file called “Zone.Identifier” which can be accessed as `filename.exe:Zone.Identifier`. You can see this alternate stream being displayed by the directory listing using the `dir /r` command:



Alternate data streams of files being listed by the “dir” command.

You can inspect this data stream by opening it in notepad from the command line. Just type `notepad filename.extension:Zone.Identifier`.

Back to our exploit. Each file contains one default data stream that contains the actual contents of the file, the ones we see when we double click and open it, or execute it. This stream is called “\$DATA”.

When we access a file hosted on a Kolibri+ Web Server 2 after appending it with `::$DATA` the web server will dutifully fetch this file and return it to you. Normally if you had a PHP or ASP file, the server would execute the code before sending it to the user requesting it. If you are fetching a PHP or ASP file as given in the exploit example:

```
http://172.16.2.101/index.php::$DATA
```

The server will no longer process the code, as it sees the extension as